

Status: Accepted

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Builds on: ADR-001 - Selection of the Open-Source LMS Foundation for Syrka; ADR-002 - Syrka Domain Architecture, Bounded Contexts & System Ownership

Decision owners: Architecture / Platform Engineering / AI Engineering / Security

Scope: Canonical event language, envelope, taxonomy, event bus, event store, projections, connector mappings, AI consumption, security, privacy, versioning, testing, and governance

1. Executive Overview

Syrka is an event-driven Human Capital Infrastructure platform. Every meaningful business fact is represented as an immutable, canonical Syrka event. These events are the universal language used by the LMS boundary, identity, ontology, capability graph, inference engine, career engine, market intelligence, national vision layer, dashboards, passports, credentials, employers, governments, and cross-border workforce corridors.

The Canonical Event Architecture exists because Syrka must never be coupled to Open edX, Canvas, Moodle, Sakai, Chamilo, ILIAS, GitHub, LinkedIn, ORCID, OpenAlex, Semantic Scholar, an employer API, a government system, or any AI provider. External systems may produce signals. They do not define Syrka's facts.

Every subsystem communicates through canonical events rather than direct database access because:

1. Replaceability. Open edX can be replaced by Canvas, Moodle, a custom LMS, or a future platform by changing only the connector/anti-corruption layer.
2. Auditability. Events create a chronological, immutable record of what happened, who/what produced it, when it occurred, and which evidence supports it.
3. AI explainability. AI systems reason over canonical, provenance-rich evidence rather than opaque third-party schemas.
4. Domain ownership. Each bounded context owns its own model and projections. No service reaches into another service's database.
5. Replayability. Syrka can rebuild projections, dashboards, capability graph state, analytics, passports, and recommendations from the event store.
6. Scalability. Producers and consumers evolve independently through versioned schemas and asynchronous delivery.
7. Compliance. Consent, privacy classification, retention, jurisdiction, and lineage travel with every event.

Events protect Syrka from LMS lock-in by turning LMS-specific activity into LMS-independent facts. For example, Open edX `problem_check`, Canvas `quiz_submission`, Moodle `mod_quiz_attempt_submitted`, and Sakai assessment submission events can all become canonical `QuizCompleted` or `AssessmentSubmitted` events. Downstream intelligence systems consume only the canonical event and therefore remain unchanged when the upstream LMS changes.

2. Event Philosophy

2.1 Event

An event is an immutable record that something happened at a specific time. Events are append-only. They are not updated. If a correction is required, a new correcting event is published.

2.2 Business Event

A business event is an event that has domain meaning to Syrka. It describes a fact the business cares about, such as AssignmentSubmitted, CapabilityObserved, PassportIssued, or EmployerInterested.

2.3 Canonical Event

A canonical event is a business event expressed in Syrka's platform-independent language, envelope, schema, privacy model, and versioning rules. Canonical events are independent of any source system's internal event names or database schemas.

2.4 Observation

An observation is a captured signal that may indicate a capability, behavior, preference, risk, or market condition. Observations are not necessarily conclusions. CommitObserved, AttendanceRecorded, and DiscussionParticipated are observations.

2.5 Evidence

Evidence is an observation, artifact, assessment, credential, or verified external signal that can support a claim. Evidence must have provenance and must be referenceable. Evidence may be strong, weak, conflicting, stale, or revoked.

2.6 Provenance

Provenance describes where an event came from, which system produced it, which source record/artifact it references, what adapter/model version transformed it, and how it can be verified or replayed.

2.7 Fact

A fact is a canonical event accepted by the event platform after validation. A fact may later be superseded or corrected, but the original fact remains part of the audit history.

2.8 Projection

A projection is derived state built from events. Examples include a student's current capability summary, an institutional dashboard aggregate, or a passport read model. Projections are rebuildable.

2.9 Read Model

A read model is an optimized query representation of one or more projections. Read models are built for product surfaces and analytics. They are not sources of truth.

2.10 Distinctions

Concept	Source of truth?	Mutable?	Example	Notes
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Event	Yes, for the fact it records	No	AssignmentSubmitted	Append-only
Observation	Yes, for observed signal	No	CommitObserved	May support inference
Evidence	Yes, as support for claims	No or revocable by new event	Submission artifact	Linked to provenance

Fact	Yes	No	Accepted canonical event	Validated event
Projection	No	Rebuildable	current capability score	Derived from events
Read model	No	Rebuildable	dashboard table	Optimized for queries
Command	No event	N/A	SubmitAssignment	Request to do something, not a fact

3. Event Taxonomy

Canonical events are grouped into categories. Categories determine ownership, governance, retention defaults, sensitivity, topic naming, and consumer expectations.

Category	Purpose	Owning domain	Typical producers	Typical consumers	Examples
Identity Events	Lifecycle of people, organizations, roles, consent, federation	Identity & Access	Auth, admin apps, SSO	LMS adapter, graph, dashboards	IdentityCreated, StudentCreated, ConsentGranted
Learning Events	Course/content interaction and progress	Learning Management ACL	Open edX/Canvas/Moodle adapters	Analytics, inference, graph	LessonViewed, ModuleCompleted
Assessment Events	Assignment, quiz, rubric, grade, feedback facts	Learning Management ACL, faculty tools	LMS adapters, assessment tools	Inference, analytics, graph	AssignmentSubmitted, QuizCompleted
Collaboration Events	Discussions, peer review, teams, attendance, group work	Learning Management ACL, collaboration tools	LMS, LTI tools	Analytics, inference	DiscussionParticipated, PeerReviewCompleted
Research Events	Publications, research projects, grants, collaborations	Research Intelligence	OpenAlex, ORCID, Semantic Scholar	Graph, portfolio, opportunity	ResearchPublished
Engineering Evidence Events	Code/project signals	GitHub Intelligence	GitHub webhooks/API	Inference, graph, portfolio	GitHubRepositoryLinked, CommitObserved
Professional Profile Events	Professional/social career evidence	LinkedIn Intelligence	LinkedIn/export imports	Career, graph	LinkedInProfileLinked
Ontology Events	Semantic model and taxonomy changes	Syrka Ontology	Ontology admin/ingestion	Inference, graph, curriculum	OntologyVersionPublished
Capability Events	Capability observations and graph updates	Inference, Graph	AI/rules/graph services	Dashboards, passport, opportunity	CapabilityObserved, CapabilityUpdated
Career Events	User intent, preferences, constraints, aspirations	Career Intent	Student app, advisor app	Opportunity, mobility	CareerIntentUpdated
Market Events	Labour-market and employer demand signals	Market Signal, Employer	Job boards, employers	Opportunity, curriculum, national	MarketSignalObserved, JobPosted
Government Events	National strategies and policy priorities	National Vision	Ministry apps, policy ingestion	Curriculum, mobility, corridors	NationalPriorityMapped
Portfolio Events	Portfolio creation, sharing, revocation	Portfolio Generator	Portfolio service	Public profile, employer	PortfolioGenerated
Passport Events	Job passport lifecycle and disclosure	Job Passport	Passport service	Employer, mobility, public profile	PassportIssued, PassportUpdated
Credential Events	Issuance, verification, revocation	Credential Ledger	Issuers, ledger service	Passport, graph, employer	CredentialVerified
Employer Events	Employer lifecycle, jobs, interest, feedback	Employer Platform	Employer portal/API	Opportunity, mobility	EmployerInterested
Mobility Events	Mobility assessment, approvals, corridor readiness	Mobility Engine	Mobility service	Corridors, government	MobilityApproved
Corridor Events	Cross-border corridor setup and matching	Syrka Corridors	Corridor service	Government, employer	CorridorMatched
Analytics Events	Risk, mastery, engagement, quality signals	AI-native Learning Analytics	Analytics workers	Faculty, institution, inference	LearningRiskDetected
AI Events	Model runs, decisions, reviews, evaluations	AI Platform	Inference/recommendation services	Audit, graph, dashboards	InferenceRunCompleted
Recommendation Events	Recommendations and feedback	Opportunity/Odyssey	Recommenders	Dashboards, audit	RecommendationGenerated

Notification Events	Delivery lifecycle of user communications	Notification service	Email/SMS/in-app service	Audit, UX	NotificationDelivered
Audit Events	Security, admin, policy, data access	Platform/Security	All services	Compliance/audit	DataAccessed, PolicyDenied
System Events	Operational lifecycle, schema, replay, health	Platform	Event platform, services	SRE, governance	SchemaPublished, ReplayCompleted

4. Event Catalogue

4.1 Catalogue format

Each event is documented with a compact implementation row. Full universal requirements are defined in sections 5 and 6 and apply to every event unless overridden.

Legend:

- Producer is the authoritative producer of the canonical event.
- Consumers are primary consumers, not an exhaustive subscription list.
- Privacy defaults: public, internal, restricted, highly_restricted.
- Retention defaults: short, standard, long, permanent, jurisdictional.

4.2 Identity events

Event	Purpose/business meaning	Producer	Consumers	Required payload fields	Optional fields	Validation	Privacy	Retention	Replay/idempotency
IdentityCreated	A platform identity was created	Identity	all domain provisioners	identity_id, identity_type, tenant_id	external_ids, unique identity_id, tenant valid	restricted permanent replay	provisions read models	idempotency by identity_id	
IdentityUpdated	Identity attributes changed	Identity	dashboards, ACLs	identity_id, changed_fields	previous_hash	identity exists; allowed fields	restricted permanent	latest projection rebuilt by order	
StudentCreated	A learner identity was created	Identity	LMS adapter, graph, analytics	student_id, identity_id, institution_id	program_id, cohort_id	identity exists; role student	restricted permanent	idempotency by student_id	
StudentUpdated	Learner profile changed	Identity	career, dashboards	student_id, changed_fields	profile_completeness	student exists	restricted permanent	replay updates profile projection	
StudentEnrolled	Student was enrolled in a learning offering	Learning Adapter	analytics, graph, faculty	student_id, course_id, enrollment_id	section_id, cohort_id	course/student mapped	restricted long	idempotency by enrollment_id	
RoleAssigned	Role granted to identity	Identity	all authz consumers	identity_id, role, scope	expires_at	role/scope valid	restricted permanent	authorization projections replayed	
RoleRevoked	Role removed	Identity	all authz consumers	identity_id, role, scope	reason	assignment exists	restricted permanent	revocation ordered after assignment	
ConsentGranted	User granted data processing/sharing consent	Identity/Consent	connectors, public profile	subject_id, consent_type, scope	expires_at	consent text/version valid	highly_restricted jurisdictional	idempotency by consent grant id	
ConsentRevoked	User revoked consent	Identity/Consent	connectors, profile, graph	subject_id, consent_type, scope	effective_at	prior grant exists	highly_restricted jurisdictional	consumers stop future processing	
FederationLinked	External identity provider linked	Identity	LMS adapter, institutions	identity_id, provider, external_subject	assurance_level	provider trusted	highly_restricted permanent	idempotency by provider subject	

4.3 Learning events

| Event | Purpose/business meaning | Producer | Consumers | Required payload fields | Optional fields | Validation
 | Privacy | Retention | Replay/idempotency |
 |--|---|---|---|---|---|---|---|
 | CourseCreated | Canonical course projection exists | Learning Adapter | curriculum, analytics | course_id,
 title, source_system | start_at, end_at | mapped source id unique | internal | long | idempotency by
 course_id/source |
 | CourseUpdated | Course metadata changed | Learning Adapter | curriculum, dashboards | course_id, changed_fields
 | syllabus_ref | course exists | internal | long | rebuild course projection |
 | ModuleCompleted | Student completed module/unit | Learning Adapter | analytics, inference | student_id,
 course_id, module_id, completed_at | duration_seconds | enrollment exists | restricted | long | idempotency by
 student+module+attempt |
 | LessonViewed | Student viewed lesson/content | Learning Adapter | analytics | student_id, lesson_id, viewed_at |
 duration_seconds, device_type | content mapped | restricted | standard | high-volume compaction allowed |
 | ContentDownloaded | Student downloaded course content | Learning Adapter | analytics, audit | student_id,
 content_id, downloaded_at | file_type | content exists | restricted | standard | idempotency source event id |
 | LearningSessionStarted | Learning session started | Learning Adapter/Web | analytics | student_id, session_id,
 started_at | device, ip_region | session unique | restricted | standard | session ordered by timestamp |
 | LearningSessionEnded | Learning session ended | Learning Adapter/Web | analytics | student_id, session_id,
 ended_at | duration_seconds | start exists or inferred | restricted | standard | updates session projection |
 | AttendanceRecorded | Attendance/participation recorded | LMS/Attendance Adapter | faculty, analytics, graph |
 student_id, activity_id, status, recorded_at | location, method | valid status | restricted | long | idempotency
 by attendance record |

4.4 Assessment and collaboration events

| Event | Purpose/business meaning | Producer | Consumers | Required payload fields | Optional fields | Validation
 | Privacy | Retention | Replay/idempotency |
 |--|---|---|---|---|---|---|---|
 | AssignmentOpened | Student opened an assignment | Learning Adapter | analytics | student_id, assignment_id,
 opened_at | attempt_id | assignment mapped | restricted | standard | high-volume event; source id |
 | AssignmentSubmitted | Student submitted assignment evidence | Learning Adapter/LTI | inference, graph, analytics
 | student_id, assignment_id, submission_id, submitted_at, artifact_refs | attempt_number, word_count | artifact
 exists; enrollment valid | highly_restricted | long | idempotency by submission_id |
 | AssignmentGraded | Assignment received grade/rubric | Learning Adapter/Faculty | inference, graph, faculty |
 submission_id, grade, graded_at, assessor_id | rubric_scores, feedback_ref | submission mapped; grade valid |
 highly_restricted | long | ordering after submission preferred |
 | QuizStarted | Quiz attempt started | Learning Adapter | analytics | student_id, quiz_id, attempt_id, started_at
 | time_limit | attempt unique | restricted | standard | idempotency by attempt_id |
 | QuizCompleted | Quiz attempt completed | Learning Adapter | inference, analytics | student_id, quiz_id,
 attempt_id, completed_at, score | item_summary, duration | attempt exists; score range valid | highly_restricted |
 long | idempotency by attempt_id+completed |
 | DiscussionCreated | Discussion topic created | Learning Adapter | analytics, faculty | discussion_id, course_id,
 creator_id | title | course mapped | restricted | standard | source event id |
 | DiscussionParticipated | User posted/replied/reacted | Learning Adapter | analytics, inference | participant_id,
 discussion_id, action, occurred_at | content_ref, sentiment_features | action valid | highly_restricted |
 standard | content minimized |
 | PeerReviewCompleted | Peer review submitted | Learning Adapter/LTI | inference, faculty | reviewer_id,
 target_submission_id, review_id | rubric_scores, feedback_ref | reviewer authorized | highly_restricted | long |
 idempotency review_id |
 | FeedbackReceived | Learner received feedback | Learning Adapter/Faculty | dashboard, inference | recipient_id,
 feedback_id, source_type | feedback_ref | recipient exists | highly_restricted | long | idempotency feedback_id |
 | ProjectStarted | Project work began | Learning/Portfolio/GitHub | graph, portfolio | project_id, owner_id,
 started_at | collaborators, repo_id | owner valid | restricted | long | idempotency project_id |

| ProjectCompleted | Project reached completion | Learning/Portfolio/GitHub | inference, graph, portfolio |
project_id, owner_id, completed_at, artifact_refs | collaborators, rubric | artifacts exist | highly_restricted |
long | idempotency project_id+version |

4.5 Research and engineering evidence events

| Event | Purpose/business meaning | Producer | Consumers | Required payload fields | Optional fields | Validation
Privacy	Retention	Replay/idempotency	
ResearchPublished	Research output published/imported	Research Intelligence	graph, portfolio, opportunity
research_id, title, authors, published_at	doi, abstract_ref, topics	source trusted or user verified	
internal/restricted	permanent	idempotency DOI/source id	
ResearchCapabilityMapped	Research mapped to capabilities	Research Intelligence	graph, dashboard
research_id, capability_ids, confidence	model_version	ontology version valid	restricted
ontology versions			
GitHubRepositoryLinked	User linked repository as evidence source	GitHub Intelligence	inference, graph
user_id, repository_id, provider, linked_at	visibility, default_branch	consent valid; ownership verified	
restricted	jurisdictional	idempotency provider repo id	
CommitObserved	Commit/contribution observed	GitHub Intelligence	inference, graph
commit_sha, author_id, committed_at	files_changed, languages	repository linked	restricted
idempotency commit_sha			
PullRequestMerged	PR merged as collaboration/code signal	GitHub Intelligence	inference, graph
repository_id, pull_request_id, merged_at, author_id	reviewers, checks	repo linked; merged true	restricted
standard	idempotency PR id+merged		
CodeReviewCompleted	Code review completed	GitHub Intelligence	inference, graph
reviewer_id, submitted_at	outcome	repo linked	restricted
LinkedInProfileLinked	User linked professional profile	LinkedIn Intelligence	career, graph
provider_profile_id, linked_at	headline, consent_scope	consent valid	highly_restricted
idempotency provider profile			
ProfessionalExperienceObserved	Experience/certification imported	LinkedIn Intelligence	career, graph
user_id, experience_id, organization, role | dates, description_ref | consent valid | highly_restricted |
jurisdictional | revoke on consent |

4.6 Ontology, capability, AI, recommendation events

| Event | Purpose/business meaning | Producer | Consumers | Required payload fields | Optional fields | Validation
| Privacy | Retention | Replay/idempotency |
|---|---|---|---|---|---|---|
| OntologyTermCreated | New semantic term created | Ontology | inference, graph, curriculum | term_id, term_type,
label, version | description | unique in version | internal | permanent | idempotency term_id+version |
| OntologyVersionPublished | Ontology version became active | Ontology | all intelligence domains | version,
published_at, changeset_ref | migration_notes | version monotonic | internal | permanent | triggers
remapping/replay |
| SkillObserved | Evidence suggests a skill | Inference/Analytics | graph, dashboard | subject_id, skill_id,
evidence_refs, confidence | explanation | ontology version valid | restricted | long | idempotency evidence+skill
|
| CapabilityObserved | Evidence supports capability claim | Inference | graph, passport, dashboard | subject_id,
capability_id, evidence_refs, confidence, model_version | explanation_ref | confidence range valid | restricted |
permanent | idempotency evidence+capability+model |
| CapabilityUpdated | Capability graph state changed | Graph | dashboard, opportunity, passport | subject_id,
capability_id, previous_state, new_state | confidence_delta | graph transition valid | restricted | permanent |
ordered per subject+capability |
| InferenceRunStarted | AI/rules inference run began | Inference | audit, ops | run_id, model_version,
input_event_ids | purpose | authorized purpose | restricted | standard | idempotency run_id |

InferenceRunCompleted	AI/rules inference run completed	Inference	audit, graph	run_id, output_event_ids, metrics	cost, latency	run exists	restricted	standard	idempotency run_id
InferenceReviewRequired	Human review required	Inference	review queue	run_id, subject_id, reason	risk_flags	high-impact rules valid	highly_restricted	long	idempotency run_id+reason
RecommendationGenerated	Recommendation produced	Opportunity/Odyssey	dashboard, audit	recommendation_id, subject_id, type, score, reasons	alternatives	model/policy valid	restricted	long	idempotency recommendation_id
RecommendationAccepted	User accepted recommendation	Product app	opportunity, analytics	recommendation_id, subject_id, accepted_at	action	recommendation exists	restricted	standard	idempotency recommendation+action

4.7 Career, market, employer, opportunity events

| Event | Purpose/business meaning | Producer | Consumers | Required payload fields | Optional fields | Validation | Privacy | Retention | Replay/idempotency |
|---|---|---|---|---|---|---|---|---|
| CareerIntentUpdated | User career goals/preferences changed | Career Intent | opportunity, mobility | subject_id, intent_id, changed_fields | salary, geography, industry | user authorized | highly_restricted | jurisdictional | latest projection replayed |
| MarketSignalObserved | Labour-market signal captured | Market Signal | opportunity, curriculum, national | signal_id, signal_type, source, observed_at | geography, confidence | source recorded | internal | long | idempotency source signal id |
| SkillDemandChanged | Skill demand materially changed | Market Signal | curriculum, opportunity | skill_id, geography, change_metric | forecast_window | ontology valid | internal | long | ordered by skill/geography |
| JobPosted | Employer published job/opportunity | Employer | opportunity, market | job_id, employer_id, title, location | salary, skills | employer verified | internal | standard | idempotency job_id |
| JobUpdated | Job details changed | Employer | opportunity | job_id, changed_fields | close_at | job exists | internal | standard | projection update |
| EmployerInterested | Employer expressed interest in candidate/profile | Employer | opportunity, mobility, audit | employer_id, subject_id, context_id, occurred_at | message_ref | consent/disclosure valid | highly_restricted | jurisdictional | idempotency interest_id |
| OpportunityMatched | Subject matched to opportunity | Opportunity | dashboard, employer, mobility | match_id, subject_id, opportunity_id, score, reasons | constraints | opportunity valid | restricted | long | idempotency match_id |
| OpportunityAccepted | Subject accepted opportunity | Product app | employer, mobility | match_id, subject_id, accepted_at | notes | match exists | restricted | standard | ordered after match |

4.8 Portfolio, passport, credential, mobility, corridor events

| Event | Purpose/business meaning | Producer | Consumers | Required payload fields | Optional fields | Validation | Privacy | Retention | Replay/idempotency |
|---|---|---|---|---|---|---|---|---|
| PortfolioGenerated | Evidence-backed portfolio created | Portfolio | public profile, employer | portfolio_id, subject_id, generated_at, evidence_refs | template_id | evidence accessible | restricted | long | idempotency portfolio version |
| PortfolioShared | Portfolio shared with audience | Portfolio | audit, employer | portfolio_id, share_id, audience, shared_at | expires_at | consent valid | highly_restricted | jurisdictional | idempotency share_id |
| CredentialIssued | Credential issued by trusted issuer | Credential Ledger | graph, passport | credential_id, issuer_id, subject_id, issued_at | vc_ref | issuer trusted | restricted | permanent | idempotency credential_id |
| CredentialVerified | Credential verification succeeded/failed | Credential Ledger | graph, passport, employer | credential_id, verification_status, verified_at | verifier_id | credential exists | restricted | permanent | verification events append-only |
| CredentialRevoked | Credential revoked | Credential Ledger | graph, passport, employer | credential_id, revoked_at, reason | issuer_signature | issuer authorized | restricted | permanent | ordered after issue |

| PassportIssued | Job Passport issued | Job Passport | employer, mobility, public profile | passport_id, subject_id, version, issued_at, claim_refs | disclosure_policy | claims verified | highly_restricted | permanent | idempotency passport version |

| PassportUpdated | Passport version/claims updated | Job Passport | employer, mobility, public profile | passport_id, version, changed_claims | reason | version monotonic | highly_restricted | permanent | ordered by passport version |

| PassportRevoked | Passport revoked or disabled | Job Passport | employer, mobility, public profile | passport_id, revoked_at, reason | replacement_id | owner authorized | highly_restricted | permanent | invalidates shares |

| MobilityApproved | Mobility assessment approved | Mobility | corridors, passport, government | mobility_case_id, subject_id, route_id, approved_at | conditions, expiry | eligibility rules satisfied | highly_restricted | permanent | idempotency case+decision |

| MobilityBlocked | Mobility assessment blocked | Mobility | dashboard, government | mobility_case_id, subject_id, reasons | remediation | case exists | highly_restricted | long | append-only decision |

| CorridorMatched | Candidate matched to corridor | Corridors | government, employer, mobility | corridor_id, subject_id, match_id, matched_at | stage | corridor active | highly_restricted | long | idempotency corridor match |

| CorridorOutcomeReported | Corridor outcome reported | Corridors | government, analytics | corridor_id, outcome_type, reported_at | metrics | partner authorized | restricted | permanent | aggregate replayable |

4.9 Analytics, notification, audit, system events

| Event | Purpose/business meaning | Producer | Consumers | Required payload fields | Optional fields | Validation | Privacy | Retention | Replay/idempotency |

---|---|---|---|---|---|---|---|---|

| DashboardViewed | User viewed dashboard/insight | Product app | analytics, audit | viewer_id, dashboard_id, viewed_at | filters | viewer authorized | restricted | short | high-volume compaction |

| LearningRiskDetected | Analytics detected learning risk | Analytics | faculty, dashboard | subject_id, risk_type, confidence, evidence_refs | recommended_actions | threshold met | highly_restricted | long | idempotency subject+risk+window |

| MasterySignalDetected | Analytics detected mastery signal | Analytics | inference, faculty | subject_id, concept_id, confidence | evidence_refs | ontology valid | restricted | long | idempotency subject+concept+window |

| NotificationRequested | Domain requested notification | Any domain | notification service | notification_id, recipient_id, channel, template | context | recipient allowed | restricted | standard | idempotency notification_id |

| NotificationDelivered | Notification delivered | Notification | audit, product | notification_id, delivered_at, channel | provider_id | request exists | restricted | standard | provider idempotency |

| DataAccessed | Sensitive data accessed | Platform/Security | audit | actor_id, resource_type, resource_id, purpose | query_hash | authorized access | highly_restricted | permanent | append-only audit |

| PolicyDenied | Authorization/policy denied action | Platform/Security | security/audit | actor_id, action, resource, reason | risk_score | policy engine result | restricted | long | append-only audit |

| SchemaPublished | Event schema published | Event Governance | all producers/consumers | schema_name, version, published_at | compatibility | approvals complete | internal | permanent | idempotency schema+version |

| ReplayStarted | Replay job started | Event Platform | SRE/audit | replay_id, event_range, consumers | reason | authorized job | internal | long | idempotency replay_id |

| ReplayCompleted | Replay job completed | Event Platform | SRE/audit | replay_id, status, counts | errors_ref | replay exists | internal | long | append-only |

| DeadLetterRecorded | Event moved to DLQ | Event Platform | SRE/domain owner | event_id, consumer, error_code | payload_ref | failure classified | restricted | long | idempotency event+consumer |

5. Event Specification

5.1 Mandatory specification fields

Every event schema must define:

1. Purpose - why the event exists.
2. Business meaning - the fact it records.
3. Producer - exactly one canonical producer domain.
4. Consumers - primary expected consumers.
5. Required fields - payload fields required in addition to the universal envelope.
6. Optional fields - additive payload fields.
7. Validation rules - schema, domain, authorization, and referential checks.
8. Privacy classification - public/internal/restricted/highly restricted.
9. Security requirements - signing, encryption, scopes, tenant boundaries.
10. Versioning rules - compatibility and migration requirements.
11. Retention policy - short/standard/long/permanent/jurisdictional.
12. Replay policy - whether replay is safe, constrained, or forbidden.
13. Example payload - canonical JSON example.
14. Failure handling - validation, retry, DLQ, quarantine.
15. Idempotency strategy - deterministic keys and duplicate handling.
16. Ordering guarantees - partition key and expected order.

5.2 Default validation rules

- Envelope validates against the universal JSON Schema.
- event_type must match the schema name.
- event_version must be registered.
- tenant_id, source_system, producer, occurred_at, and idempotency_key are required.
- actor, subject, and resource must use canonical ID formats where applicable.
- Required payload fields must be present and type-valid.
- Payload must not contain raw secrets, passwords, tokens, or unclassified large artifacts.
- Evidence artifacts must be referenced through secure evidence_refs, not embedded.
- Events from connectors must include source provenance.

5.3 Default privacy classifications

| Classification | Meaning | Examples | Handling |

|---|---|---|---|

| public | Intended for public disclosure | published public profile event | signed; no sensitive data |

| internal | Syrka internal operational data | schema published, market aggregate | tenant-aware access |

| restricted | Personal, institutional, or business-sensitive | course progress, job matches | encrypted, scoped authorization |

| highly_restricted | High-impact personal/evidence data | grades, passport, consent, mobility | strict authorization, audit, minimization |

5.4 Example event payload - AssignmentSubmitted

```
{
  "event_id": "01JZCEA00000000000000000001",
  "event_type": "AssignmentSubmitted",
  "event_version": "1.0.0",
  "schema_version": "1.0.0",
  "occurred_at": "2026-07-07T10:15:00Z",
  "published_at": "2026-07-07T10:15:03Z",
  "correlation_id": "01JZCORR000000000000000001",
  "causation_id": "01JZCAUSE000000000000000001",
```

```

"trace_id": "00-4bf92f3577b34da6a3ce929d0e0e4736-00f067aa0ba902b7-00",
"tenant_id": "tenant_sa_001",
"institution_id": "inst_ksu_001",
"country_code": "SA",
"source_system": "openedx",
"producer": "learning-management-adapter",
"actor": { "type": "student", "id": "stu_123" },
"subject": { "type": "student", "id": "stu_123" },
"resource": { "type": "assignment", "id": "assign_456" },
"evidence_refs": [
  { "type": "object", "uri": "s3://syrra-evidence/tenant_sa_001/submissions/sub_789.pdf", "hash": "sha256:..." }
],
"privacy": { "classification": "highly_restricted", "consent_basis": "education", "jurisdiction": "SA" },
"security": { "signature": "base64url-jws", "signature_alg": "EdDSA", "key_id": "kid_2026_01" },
"metadata": { "adapter_version": "openedx-acl@1.4.0" },
"payload": {
  "submission_id": "sub_789",
  "assignment_id": "assign_456",
  "course_id": "course_101",
  "attempt_number": 1,
  "submitted_at": "2026-07-07T10:15:00Z",
  "artifact_count": 1
},
"provenance": {
  "source_event_id": "openedx:event:abc123",
  "source_record_id": "openedx_submission_987",
  "source_event_type": "submission_created",
  "transformer": "openedx-assignment-submission-mapper",
  "transformer_version": "1.4.0"
},
"idempotency_key": "openedx:openedx_submission_987:AssignmentSubmitted:1.0.0"
}

```

5.5 Failure handling defaults

- Producer validation failure: reject before publish and log EventRejected internally.
- Broker publish failure: retry with exponential backoff and jitter.
- Consumer transient failure: retry according to consumer policy.
- Consumer permanent failure: move to DLQ with error classification.
- Privacy/security failure: quarantine, alert Security, and block replay until reviewed.
- Schema mismatch: reject and notify schema owner and producer owner.

5.6 Ordering guarantees

Syrka provides ordering within a partition key, not globally. Recommended partition keys:

```

| Event family | Partition key |
|---|---|
| Identity | identity_id |
| Learning/assessment | student_id or student_id:course_id |
| Capability | subject_id:capability_id |
| Passport | passport_id |
| Credential | credential_id |

```

Mobility	mobility_case_id
Corridor	corridor_id
Market	geography:occupation_or_skill_id

6. Universal Event Envelope

6.1 Required envelope fields

Field	Required	Meaning
event_id	yes	Globally unique immutable event identifier, preferably UUIDv7/ULID
event_type	yes	Canonical PascalCase business event name
event_version	yes	Version of this event type
schema_version	yes	Envelope/schema registry version
occurred_at	yes	Time the business fact occurred
published_at	yes	Time event was accepted/published
correlation_id	yes	Groups related workflow events
causation_id	no	Event/command that caused this event
trace_id	yes	Distributed tracing ID
tenant_id	yes	Tenant boundary
institution_id	conditional	Institution when applicable
country_code	conditional	ISO country/jurisdiction when applicable
source_system	yes	Original source, e.g. openedx, canvas, syrka
producer	yes	Canonical producer service/domain
actor	conditional	Entity that performed action
subject	conditional	Entity the event is about
resource	conditional	Entity/resource acted upon
evidence_refs	no	Evidence artifact references
privacy	yes	Classification, consent, jurisdiction
security	yes	Signature and verification metadata
metadata	no	Non-domain operational metadata
payload	yes	Event-specific body
provenance	yes	Source and transformation details
idempotency_key	yes	Deterministic duplicate detection key

6.2 Canonical JSON Schema - envelope

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://schemas.syrka.ai/events/envelope/1.0.0",
  "title": "SyrkaCanonicalEventEnvelope",
  "type": "object",
  "required": [
    "event_id", "event_type", "event_version", "schema_version",
    "occurred_at", "published_at", "correlation_id", "trace_id",
    "tenant_id", "source_system", "producer", "privacy", "security",
    "payload", "provenance", "idempotency_key"
  ],
  "properties": {
    "event_id": { "type": "string", "minLength": 10 },
    "event_type": { "type": "string", "pattern": "^[A-Z][A-Za-z0-9]+$" },
    "event_version": { "type": "string", "pattern": "^[0-9]+\\.[0-9]+\\.[0-9]+$" },
    "schema_version": { "type": "string", "pattern": "^[0-9]+\\.[0-9]+\\.[0-9]+$" },
  }
}
```

```

"occurred_at": { "type": "string", "format": "date-time" },
"published_at": { "type": "string", "format": "date-time" },
"correlation_id": { "type": "string" },
"causation_id": { "type": ["string", "null"] },
"trace_id": { "type": "string" },
"tenant_id": { "type": "string" },
"institution_id": { "type": ["string", "null"] },
"country_code": { "type": ["string", "null"], "pattern": "^[A-Z]{2}$" },
"source_system": { "type": "string" },
"producer": { "type": "string" },
"actor": { "$ref": "#/$defs/entityRef" },
"subject": { "$ref": "#/$defs/entityRef" },
"resource": { "$ref": "#/$defs/entityRef" },
"evidence_refs": { "type": "array", "items": { "$ref": "#/$defs/evidenceRef" } },
"privacy": { "$ref": "#/$defs/privacy" },
"security": { "$ref": "#/$defs/security" },
"metadata": { "type": "object", "additionalProperties": true },
"payload": { "type": "object" },
"provenance": { "$ref": "#/$defs/provenance" },
"idempotency_key": { "type": "string", "minLength": 8 }
},
"$defs": {
  "entityRef": {
    "type": "object",
    "required": ["type", "id"],
    "properties": {
      "type": { "type": "string" },
      "id": { "type": "string" }
    },
  },
  "additionalProperties": false
},
"evidenceRef": {
  "type": "object",
  "required": ["type", "uri"],
  "properties": {
    "type": { "type": "string" },
    "uri": { "type": "string" },
    "hash": { "type": "string" },
    "content_type": { "type": "string" },
    "created_at": { "type": "string", "format": "date-time" }
  }
},
"privacy": {
  "type": "object",
  "required": ["classification"],
  "properties": {
    "classification": { "enum": ["public", "internal", "restricted", "highly_restricted"] },
    "consent_basis": { "type": ["string", "null"] },
    "jurisdiction": { "type": ["string", "null"] },
    "retention_policy": { "type": ["string", "null"] }
  }
},
"security": {
  "type": "object",

```

```

"required": ["signature", "signature_alg", "key_id"],
"properties": {
  "signature": { "type": "string" },
  "signature_alg": { "type": "string" },
  "key_id": { "type": "string" }
}
},
"provenance": {
  "type": "object",
  "required": ["source_event_type", "transformer", "transformer_version"],
  "properties": {
    "source_event_id": { "type": ["string", "null"] },
    "source_record_id": { "type": ["string", "null"] },
    "source_event_type": { "type": "string" },
    "source_url": { "type": ["string", "null"] },
    "transformer": { "type": "string" },
    "transformer_version": { "type": "string" }
  }
}
}
}
}

```

7. Event Relationships

7.1 Learning-to-mobility chain

AssignmentSubmitted

v

AssignmentGraded

v

CapabilityObserved

v

CapabilityUpdated

v

RecommendationGenerated

v

PassportUpdated

v

OpportunityMatched

v

EmployerInterested

v

MobilityApproved

7.2 Sequence diagram

sequenceDiagram

participant LMS as LMS / Learning Tool

participant ACL as Connector ACL

participant Bus as Canonical Event Bus

participant AI as Capability Inference
 participant Graph as Capability Graph
 participant Rec as Opportunity Engine
 participant Pass as Job Passport
 participant Emp as Employer Platform
 participant Mob as Mobility Engine
 LMS->>ACL: Native submission/grade event
 ACL->>Bus: AssignmentSubmitted
 ACL->>Bus: AssignmentGraded
 Bus->>AI: Deliver assessment evidence
 AI->>Bus: CapabilityObserved
 Bus->>Graph: Update capability state
 Graph->>Bus: CapabilityUpdated
 Bus->>Rec: Trigger recommendation update
 Rec->>Bus: RecommendationGenerated
 Bus->>Pass: Refresh passport claims
 Pass->>Bus: PassportUpdated
 Bus->>Rec: Match opportunities
 Rec->>Bus: OpportunityMatched
 Emp->>Bus: EmployerInterested
 Bus->>Mob: Evaluate mobility case
 Mob->>Bus: MobilityApproved

7.3 Causation and correlation rules

- correlation_id remains constant across a business workflow.
- causation_id points to the event that directly caused another event.
- trace_id follows runtime request/worker execution.
- Capability events must include evidence event IDs.
- Recommendation and passport events must include upstream capability event IDs or graph version references.

8. Canonical Event Mapping - LMS Platforms

Mapping is owned by the connector anti-corruption layer. Downstream systems never consume native LMS events.

8.1 Open edX mapping

Native Open edX signal	Canonical event	Notes
enrollment created	StudentEnrolled	Map course run and learner IDs
course published/imported	CourseCreated / CourseUpdated	Studio metadata to course projection
sequential/vertical completed	ModuleCompleted	Normalize block IDs
page/block viewed	LessonViewed	High-volume analytics event
problem_check	QuizCompleted or AssessmentSubmitted	Depends on assessment type
submission_created	AssignmentSubmitted	Artifact refs required
score_changed / grade updated	AssignmentGraded	Include rubric if available
discussion post/reply	DiscussionParticipated	Content stored by reference
certificate generated	CredentialIssued candidate	Verified by Credential Ledger before CredentialVerified
team activity	ProjectStarted / ProjectCompleted	Requires project mapping

8.2 Canvas mapping

Native Canvas signal	Canonical event	Notes
user/account created	IdentityCreated candidate or StudentCreated via Identity	Identity remains Syrka-owned
enrollment_created	StudentEnrolled	Section/course mapped
course_created	CourseCreated	Canvas course ID mapped
module_item_completed	ModuleCompleted	Module progress
page_view	LessonViewed	High-volume compactable
assignment_created	CourseUpdated or curriculum event	Assignment entity projection
submission_created	AssignmentSubmitted	Submission artifact refs
submission_graded	AssignmentGraded	Grade/rubric/feedback
quiz_submission_created	QuizStarted	Attempt begins
quiz_submission_completed	QuizCompleted	Attempt completes
discussion_entry_created	DiscussionParticipated	Post/reply action

8.3 Moodle mapping

Native Moodle signal	Canonical event	Notes
user_created	IdentityCreated candidate	Prefer Syrka Identity as source
role_assigned	RoleAssigned candidate	Tenant scoped
user_enrolment_created	StudentEnrolled	Map enrolment plugin source
course_created	CourseCreated	Moodle course ID mapped
course_module_completion_updated	ModuleCompleted	Activity completion
course_viewed	LessonViewed	Context determines resource
mod_assign_submission_created	AssignmentSubmitted	Normalize files/text
assign_submission_graded	AssignmentGraded	Gradebook mapping
mod_quiz_attempt_started	QuizStarted	Attempt ID required
mod_quiz_attempt_submitted	QuizCompleted	Score may arrive separately
mod_forum_post_created	DiscussionParticipated	Content by reference

8.4 Sakai mapping

Native Sakai signal	Canonical event	Notes
site membership added	StudentEnrolled	Site/course mapping
content resource viewed	LessonViewed	Resource ID mapped
assignment submission	AssignmentSubmitted	Submission refs
assignment grade released	AssignmentGraded	Gradebook integration
assessment started	QuizStarted	Samigo/tests mapping
assessment submitted	QuizCompleted	Scores normalized
forum message created	DiscussionParticipated	Topic/reply action
calendar/attendance event	AttendanceRecorded	Tool-specific mapping

8.5 Chamilo mapping

Native Chamilo signal	Canonical event	Notes
course subscription	StudentEnrolled	User/course mapped
learning path item completed	ModuleCompleted	Learning path normalized
document viewed/downloaded	LessonViewed / ContentDownloaded	Content type determines event
assignment submitted	AssignmentSubmitted	Artifact refs
exercise attempt started	QuizStarted	Attempt ID synthesized if needed
exercise completed	QuizCompleted	Score normalized

| forum post created | DiscussionParticipated | Content by reference |
| attendance entry | AttendanceRecorded | Status normalized |

8.6 ILIAS mapping

Native ILIAS signal	Canonical event	Notes
course membership assigned	StudentEnrolled	Role and object mapping
learning module progress	ModuleCompleted	Repository object mapped
content object viewed	LessonViewed	High-volume analytics
exercise submitted	AssignmentSubmitted	File/text refs
exercise graded	AssignmentGraded	Feedback/grade normalized
test started	QuizStarted	Test attempt ID
test finished	QuizCompleted	Score normalized
forum posting created	DiscussionParticipated	Content reference

8.7 Future LMS mapping contract

Any future LMS must implement:

1. Identity mapping to Syrka canonical identities.
2. Course/resource mapping registry.
3. Enrollment event production.
4. Learning activity event production.
5. Assessment/submission/grade event production.
6. Evidence artifact reference export.
7. Connector provenance and idempotency keys.
8. Replay/backfill support.
9. DLQ and monitoring hooks.
10. Versioned adapter contract tests.

9. External Connector Events

9.1 GitHub

GitHub signal	Canonical event	Required normalization
OAuth app installed/link approved	GitHubRepositoryLinked	consent, owner proof, repo identity
push/commit webhook	CommitObserved	author mapping, SHA, files/languages
pull_request closed+merged	PullRequestMerged	PR ID, reviewers, checks
pull_request_review submitted	CodeReviewCompleted	reviewer, outcome, comments ref
workflow_run completed	ProjectCompleted or quality signal	checks/tests as evidence

9.2 LinkedIn

LinkedIn/export signal	Canonical event	Required normalization
profile linked	LinkedInProfileLinked	consent, profile proof
experience imported	ProfessionalExperienceObserved	role, organization, dates
certification imported	CredentialIssued candidate	verified later by Credential Ledger
skill/endorsement imported	SkillObserved candidate	low confidence unless verified

9.3 ORCID

ORCID signal	Canonical event	Required normalization
ORCID linked	ProfessionalExperienceObserved or identity link event	researcher identity proof
work added	ResearchPublished	DOI/source mapping
affiliation added	ProfessionalExperienceObserved	institution mapping

9.4 OpenAlex and Semantic Scholar

Source signal	Canonical event	Required normalization
work discovered	ResearchPublished	DOI/OpenAlex/S2 IDs, authors
citation update	MarketSignalObserved or research metric projection	citations are signals, not capability facts
topic classification	ResearchCapabilityMapped	ontology version/model provenance
collaboration pattern	RecommendationGenerated candidate	research collaboration suggestions

9.5 Employer APIs

Employer API signal	Canonical event	Required normalization
job created	JobPosted	occupation/skill mapping
job updated/closed	JobUpdated	status, dates
candidate shortlisted	EmployerInterested	consent/disclosure validation
hire/feedback submitted	OpportunityAccepted or hiring feedback event	outcome and feedback refs

9.6 Government APIs

Government signal	Canonical event	Required normalization
strategy document published	NationalPriorityMapped after parsing	source document and policy refs
priority sector updated	NationalPriorityMapped	sector/occupation/capability mapping
workforce target changed	MarketSignalObserved / government event	country/jurisdiction metadata
mobility rule changed	mobility rule event	legal/policy source required

9.7 Future connectors

Future connectors must publish only canonical events, pass contract tests, support consent enforcement, provide provenance, and include replay/backfill tooling before production approval.

10. Event Bus Architecture

10.1 Broker choice

The architecture is broker-agnostic. Recommended progression:

1. Early stage: managed Postgres/Supabase queues or a managed message broker for simplicity.
2. Scale stage: Kafka-compatible broker or NATS JetStream for durable streams, partitions, replay, and consumer groups.
3. Enterprise stage: multi-region Kafka-compatible event platform with schema registry, tiered storage, and strict governance.

10.2 Topic design

Topic names use lowercase dotted domains:

syrka.identity.v1
syrka.learning.v1
syrka.assessment.v1
syrka.collaboration.v1
syrka.research.v1
syrka.engineering_evidence.v1
syrka.ontology.v1
syrka.capability.v1
syrka.career.v1
syrka.market.v1
syrka.government.v1
syrka.portfolio.v1
syrka.passport.v1
syrka.credential.v1
syrka.employer.v1
syrka.mobility.v1
syrka.corridor.v1
syrka.analytics.v1
syrka.ai.v1
syrka.recommendation.v1
syrka.notification.v1
syrka.audit.v1
syrka.system.v1

10.3 Queues and consumers

- Consumers belong to named consumer groups.
- Each bounded context owns its consumer group names.
- Retry queues are per topic or per consumer group.
- DLQs preserve original event, error metadata, consumer, attempt count, and last exception class.

10.4 Partitions and ordering

Partition by stable business key, not random event ID. Partition keys are defined in section 5.6. Global ordering is not guaranteed and must not be required.

10.5 Delivery semantics

Syrka uses at-least-once delivery with idempotent producers and consumers. Exactly-once end-to-end delivery is not assumed. Exactly-once-like behavior is implemented through idempotency keys, transactional outbox, consumer checkpoints, and deterministic projections.

10.6 Retry and back-pressure

- Use exponential backoff with jitter.
- Use bounded retries before DLQ.
- Consumers must expose lag metrics.
- Producers must support back-pressure and circuit breakers.
- High-volume analytics events may be sampled only if explicitly declared and not used as

durable evidence.

10.7 Throughput and horizontal scaling

- Partition counts scale by tenant/institution volume.
- Consumers scale horizontally within consumer groups.
- Heavy AI consumers should decouple event ingestion from model execution with work queues.
- Batch replays should use isolated consumer groups to avoid starving live processing.

10.8 Observability

Required metrics:

- event publish rate
- publish latency
- validation failures
- consumer lag
- retry count
- DLQ volume
- replay throughput
- schema rejection count
- partition skew
- end-to-end workflow latency
- PII/security policy violations

11. Event Store

11.1 Purpose

The event store is the immutable source of canonical business facts. It must support audit, replay, projection rebuilds, compliance review, model reproducibility, and disaster recovery.

11.2 Storage model

Recommended layers:

1. Hot stream store: broker retention for recent operational replay.
2. Canonical event lake: append-only object storage partitioned by date/topic/tenant/country.
3. Event index: PostgreSQL index of event metadata for discovery and audit.
4. Schema registry: versioned schema definitions and compatibility metadata.

11.3 Retention

| Retention class | Default period | Examples |

|---|---:|---|

| short | 30-90 days | dashboard views, transient notifications |

| standard | 1-3 years | content views, sessions |

| long | 7-10 years | learning evidence, analytics decisions |

| permanent | indefinite unless law requires deletion/anonymization | credentials, passports, audit facts |

| jurisdictional | policy-specific | consent, minors, country-specific education records |

11.4 Snapshots and compaction

- Snapshots speed up replay for large aggregates such as capability graph and passport state.

- Snapshots are derived and replaceable, not sources of truth.
- Compactable events must be explicitly marked; evidence events are not compacted away from the canonical event lake.

11.5 Archiving

Cold events move to lower-cost storage with searchable metadata indexes. Archive format must remain open and documented: JSON Lines or Parquet with schema version metadata.

11.6 Disaster recovery and multi-region

- Event lake replicated across regions according to customer/jurisdiction policy.
- Recovery point objective target: ≤ 15 minutes for canonical events.
- Recovery time objective target: ≤ 4 hours for event ingestion core.
- Multi-region active-active requires deterministic event IDs and regional conflict policy.

12. Read Models

12.1 Projection construction

Projections are built by consumers that subscribe to canonical events and write domain-owned read models. Projection code must be deterministic, idempotent, observable, and replayable.

12.2 Dashboards

Dashboards consume read models from Graph, Analytics, Opportunity, Passport, and Institutional Analytics. Dashboard services must not recompute facts from raw LMS tables.

12.3 Analytics

Analytics projections aggregate learning events by learner, cohort, course, institution, capability, time window, and jurisdiction. Aggregates retain pointers to source event IDs.

12.4 Job Passport

The Job Passport is built from CapabilityUpdated, CredentialVerified, CareerIntentUpdated, and selected portfolio/public-profile events. Passport versions include claim references and provenance.

12.5 Capability Graph

The graph updates from CapabilityObserved, CapabilityUpdated, CredentialVerified, ResearchCapabilityMapped, CodeCapabilityEvidenceObserved-style events, and ontology version events. Every edge references evidence events.

12.6 AI projections

AI services may consume vectorized evidence projections, graph neighborhoods, and feature stores, but model outputs must reference original canonical event IDs.

13. AI Integration

13.1 AI processing pipeline

AssignmentSubmitted
v
Evidence Extraction
v
Ontology Mapping
v
Capability Inference
v
Confidence Scoring
v
Human Review when required
v
CapabilityObserved
v
CapabilityUpdated
v
Career Intelligence
v
RecommendationGenerated
v
PassportUpdated

13.2 Stages

1. Evidence extraction. Extract text, metadata, rubric scores, code metrics, citations, or artifacts from evidence references.
2. Normalization. Convert raw evidence into feature records with source event IDs.
3. Ontology mapping. Match evidence to skills, knowledge, occupations, and capabilities using ontology version metadata.
4. Retrieval. Pull relevant graph context, prior evidence, rubric definitions, and market signals.
5. Inference. Apply deterministic rules, statistical models, and LLM reasoning through provider adapters.
6. Confidence scoring. Combine evidence quality, recency, difficulty, assessor trust, model uncertainty, and corroboration.
7. Explainability. Generate evidence-linked explanations and limitations.
8. Policy and bias checks. Apply fairness, safety, and high-impact decision gates.
9. Publication. Publish CapabilityObserved, RecommendationGenerated, or review events.
10. Feedback loop. Consume user/faculty/employer feedback as new events.

13.3 AI-provider independence

AI providers are adapters. Canonical events must not include provider-specific opaque structures as required fields. Provider outputs are stored with model name, version, prompt/template ID, evaluation version, and evidence references.

14. Security Model

14.1 Encryption

- Transport encryption is mandatory for all producers, brokers, consumers, and storage.
- Sensitive event lake partitions are encrypted at rest.

- Highly restricted evidence artifacts use object-level encryption where supported.

14.2 Signing and verification

- Producers sign events or publish through infrastructure that signs accepted events.
- Consumers verify event signatures for high-impact domains.
- Key rotation is mandatory and tracked by key_id.

14.3 Replay protection

- Event IDs and idempotency keys prevent duplicate acceptance.
- Replay jobs use new replay metadata; they do not mint new business event IDs.
- Consumers distinguish live delivery from replay context.

14.4 Tamper detection

- Event lake objects include hashes/manifests.
- Append-only storage and audit logs detect unauthorized mutation.
- Credential/passport events may be anchored in external verification systems where required.

14.5 Authorization and multi-tenancy

- Producers require domain-specific publish scopes.
- Consumers require domain-specific subscribe scopes.
- Tenant boundaries are enforced at broker, consumer, projection, and API layers.
- Cross-tenant aggregation requires explicit policy and anonymization.

14.6 Audit and compliance

Every sensitive event access emits DataAccessed or equivalent audit records. Security policies must support FERPA, GDPR, local education regulations, labour-market privacy, credential law, and government data-sharing agreements.

15. Privacy Model

15.1 Consent

Consent is evented. ConsentGranted and ConsentRevoked determine whether connectors, public profiles, employer sharing, and professional data processing are allowed.

15.2 GDPR compatibility

- Data minimization is mandatory.
- Processing basis is recorded in the privacy block.
- Right to access is served through projections and event index search.
- Right to deletion is implemented through deletion/anonymization workflows where legally allowed; immutable event history may be cryptographically tombstoned or anonymized according to policy.

15.3 FERPA compatibility

- Education records are restricted or highly restricted.
- Access requires legitimate educational interest or explicit consent.
- Disclosure to employers/governments requires purpose, consent, and audit.

15.4 Jurisdiction overlays

Jurisdiction-specific policies may override retention, processing basis, data residency, cross-border transfer, and minor-student rules. The event envelope carries jurisdiction metadata.

15.5 Sensitive and public events

Sensitive events include grades, submissions, capability inferences, passports, credentials, mobility decisions, consent, identity federation, and employer interest. Public events are rare and normally represent user-published profile updates or public research metadata.

16. Event Versioning

16.1 Schema evolution

- Minor versions may add optional fields.
- Patch versions clarify validation without changing compatible structure.
- Major versions introduce breaking changes.
- Field removal is breaking.
- Semantic meaning changes are breaking even if JSON shape is unchanged.

16.2 Backward compatibility

Consumers must ignore unknown optional fields. Producers must continue publishing previous major versions during migration windows.

16.3 Breaking changes

Breaking changes require:

1. Schema proposal.
2. Impact analysis.
3. Approval by event governance board and affected domain owners.
4. Dual-publish or transform plan.
5. Replay/backfill plan.
6. Deprecation date.
7. ADR if the meaning or ownership changes.

16.4 Deprecation

Deprecated events remain documented. Producers stop publishing only after all consumers migrate and replay strategy is validated.

17. Performance Targets

Initial targets are planning estimates and must be validated during load testing.

Scale stage	Institutions	Learners	Peak events/sec	P95 publish-to-consume latency	Replay target
---	---	---	---	---	---
Pilot	1-5	1k-25k	100-500	< 5s	1 day in < 1 hour
Regional	10-100	25k-1M	1k-10k	< 10s	30 days in < 8 hours
National	100-1k	1M-20M	10k-100k	< 30s	90 days in < 24 hours
Multi-country	1k+	20M+	100k+	< 60s by priority class	partitioned replay by tenant/country

Storage growth estimate:

- Average envelope event: 2-8 KB excluding artifacts.
- High-volume view/session events dominate count.
- Evidence artifacts stored separately in object storage.
- Event lake should plan for TB-scale at national stage and PB-scale at multi-country stage.

Scaling strategy:

- Partition by tenant, subject, capability, or geography.
- Separate high-volume analytics topics from durable evidence topics.
- Use tiered storage for older events.
- Use read-model rebuild clusters for replay.
- Isolate AI processing queues to protect core ingestion.

18. Testing Strategy

| Test type | Purpose | Required coverage |

|---|---|---|

| Unit tests | Validate mappers, schema validators, idempotency code | every producer/consumer |

| Integration tests | Validate publish/consume with broker and store | every event family |

| Contract tests | Ensure producers and consumers obey schemas | every schema version |

| Replay tests | Rebuild projections from event history | graph, passport, dashboards, analytics |

| Chaos tests | Validate broker/consumer failure behavior | critical consumers |

| Load tests | Validate throughput, lag, back-pressure | scale milestones |

| Failure injection | Validate retries, DLQ, poison events | connectors and AI consumers |

| Schema validation tests | Reject invalid event payloads | CI and runtime |

| Consumer validation tests | Ensure idempotent/ordered behavior | all stateful consumers |

| Privacy/security tests | Validate classification, consent, scopes | sensitive events |

Minimum CI requirements for event producers:

1. Schema validation passes.
2. Example payloads validate.
3. Idempotency key test passes.
4. Contract tests pass against latest and supported previous schemas.
5. No forbidden fields/secrets in payload.
6. Privacy classification present.

19. Governance

19.1 Ownership

The Platform/Event Infrastructure Team owns the envelope, broker standards, schema registry, event store, and governance workflow. Each bounded context owns schemas for its event types.

19.2 Approval process

Event schema changes require:

1. Domain owner approval.
2. Platform/Event review.
3. Security/privacy review for restricted/highly restricted events.

4. AI review if events feed high-impact models.
5. Documentation update in CEA or schema registry.
6. Migration/replay plan for changes affecting existing consumers.

19.3 Breaking changes

Breaking changes require architecture review and, when domain ownership or semantic meaning changes, a new ADR.

19.4 Deprecation handling

Deprecated events are marked in the registry with replacement event, last producer date, consumer migration status, and replay transform guidance.

19.5 ADR governance

CEA-001 is the constitutional event specification. Future changes that alter fundamental principles-immutability, canonical envelope, LMS independence, ownership, privacy, or replayability-must update or supersede this document through a new accepted ADR/CEA.

20. Event Naming Convention

20.1 Mandatory rules

- Use PascalCase.
- Use past-tense facts.
- Use business language, not technical implementation language.
- Avoid source-system names.
- Avoid version numbers in names.
- Avoid vague words like Thing, Data, Event, Foo, Test.
- Name the business outcome, not the command.

20.2 Good names

- AssignmentSubmitted
- AssignmentGraded
- CapabilityObserved
- CapabilityUpdated
- ResearchPublished
- CredentialVerified
- PassportIssued
- MobilityApproved

20.3 Forbidden names

- assignment_complete_v2
- foo_event
- test123
- openedx_problem_check
- canvas_submission_created
- doRecommendation
- UpdatePassportCommand

20.4 Naming pattern

<Noun><PastTenseVerb>

Examples:

AssignmentSubmitted
QuizCompleted
GitHubRepositoryLinked
CareerIntentUpdated
OpportunityMatched

21. Mermaid Diagrams

21.1 Event Taxonomy

```
mindmap
  root((Syrka Events))
    Identity
      IdentityCreated
      StudentCreated
      ConsentGranted
    Learning
      LessonViewed
      ModuleCompleted
      AttendanceRecorded
    Assessment
      AssignmentSubmitted
      AssignmentGraded
      QuizCompleted
    Evidence
      ResearchPublished
      GitHubRepositoryLinked
      CommitObserved
    Intelligence
      CapabilityObserved
      CapabilityUpdated
      RecommendationGenerated
    Trust
      CredentialVerified
      PassportIssued
      PassportUpdated
    Workforce
      EmployerInterested
      MobilityApproved
      CorridorMatched
    Platform
      SchemaPublished
      ReplayCompleted
      DataAccessed
```

21.2 Event Lifecycle

flowchart LR

```
Source[Source system] --> Extract[Extract]
Extract --> Map[Map to canonical event]
Map --> Validate[Validate schema/domain/privacy]
Validate --> Sign[Sign/enrich envelope]
Sign --> Publish[Publish to bus]
Publish --> Store[Append to event store]
Publish --> Consume[Consumers]
Consume --> Project[Update projections]
Project --> Observe[Metrics/audit]
Validate -- invalid --> DLQ[Reject/DLQ]
```

21.3 Event Flow

flowchart TB

```
LMS[Open edX/Canvas/Moodle/etc.] --> ACL[LMS ACL]
GitHub[GitHub] --> Connectors[External Connectors]
Research[Research Sources] --> Connectors
Employer[Employer APIs] --> Connectors
ACL --> Bus[Canonical Event Bus]
Connectors --> Bus
Bus --> Store[(Immutable Event Store)]
Bus --> AI[AI/Inference Consumers]
Bus --> Graph[Capability Graph]
Bus --> Analytics[Analytics]
Bus --> Passport[Job Passport]
Bus --> Projections[(Read Models)]
```

21.4 Event Bus

flowchart LR

```
Producers[Producers] --> Validator[Schema/Policy Validator]
Validator --> Broker[Broker Topics]
Broker --> CG1[Consumer Group: Graph]
Broker --> CG2[Consumer Group: Analytics]
Broker --> CG3[Consumer Group: Inference]
Broker --> CG4[Consumer Group: Passport]
Broker --> DLQ[Dead Letter Queues]
Broker --> Lake[Event Lake]
Schema[Schema Registry] --> Validator
Observability[Metrics/Tracing] -. monitors .- Broker
```

21.5 Replay Process

sequenceDiagram

participant Operator as Authorized Operator
participant Replay as Replay Service
participant Store as Event Store
participant Bus as Replay Topic
participant Consumer as Projection Consumer
Operator->>Replay: Start replay with scope/reason
Replay->>Store: Read events by topic/time/tenant
Replay->>Bus: Publish replay context events
Bus->>Consumer: Deliver events with replay metadata
Consumer->>Consumer: Idempotent rebuild/checkpoint
Consumer->>Replay: Report progress
Replay->>Operator: ReplayCompleted

21.6 Event Store

flowchart TB

Bus[Event Bus] --> Hot[Hot Stream Retention]
Bus --> Lake[Append-only Event Lake]
Lake --> Index[(Event Metadata Index)]
Lake --> Archive[Cold Archive]
Registry[Schema Registry] --> Lake
Lake --> Replay[Replay Service]
Replay --> Bus

21.7 AI Processing Pipeline

flowchart LR

Event[Evidence Event] --> Extract[Evidence Extraction]
Extract --> Embed[Embeddings]
Extract --> OntMap[Ontology Mapping]
OntMap --> Retrieve[Graph/Vector Retrieval]
Retrieve --> Infer[Inference Engine]
Infer --> Confidence[Confidence Scoring]
Confidence --> Explain[Explanation]
Explain --> Policy[Bias/Safety/Policy]
Policy --> Publish[CapabilityObserved]

21.8 Connector Architecture

flowchart LR

Native[Native API/Webhook/Event] --> Auth[Connector Auth/Consent]
Auth --> Raw[Raw Event Capture]
Raw --> Mapper[Canonical Mapper]
Mapper --> Validate[Schema/Domain Validation]
Validate --> Idem[Idempotency Check]
Idem --> Publish[Publish Canonical Event]

Validate --> DLQ[Connector DLQ]

21.9 Projection Architecture

flowchart TB

```
Bus[Event Bus] --> Consumer[Projection Consumer]
Consumer --> Checkpoint[(Consumer Checkpoint)]
Consumer --> Projector[Deterministic Projector]
Projector --> ReadModel[(Read Model)]
Projector --> Snapshot[(Snapshot)]
ReadModel --> API[Domain API]
API --> Dashboard[Dashboard]
```

21.10 Job Passport Update Flow

sequenceDiagram

```
participant Graph as Capability Graph
participant Bus as Event Bus
participant Passport as Job Passport Service
participant Credential as Credential Ledger
participant Public as Public Profile
Graph->>Bus: CapabilityUpdated
Credential->>Bus: CredentialVerified
Bus->>Passport: Deliver capability/credential events
Passport->>Passport: Rebuild passport claims
Passport->>Bus: PassportUpdated
Bus->>Public: Update public disclosure projection if allowed
```

22. Appendices

Appendix A - Event-specific JSON Schema example: AssignmentSubmitted

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "https://schemas.syrka.ai/events/AssignmentSubmitted/1.0.0",
  "allOf": [
    { "$ref": "https://schemas.syrka.ai/events/envelope/1.0.0" },
    {
      "type": "object",
      "properties": {
        "event_type": { "const": "AssignmentSubmitted" },
        "payload": {
          "type": "object",
          "required": ["submission_id", "assignment_id", "course_id", "submitted_at"],
          "properties": {
            "submission_id": { "type": "string" },
            "assignment_id": { "type": "string" },
            "course_id": { "type": "string" },

```

```

    "attempt_number": { "type": "integer", "minimum": 1 },
    "submitted_at": { "type": "string", "format": "date-time" },
    "artifact_count": { "type": "integer", "minimum": 0 }
  },
  "additionalProperties": false
}
}
}
]
}

```

Appendix B - Example event stream

```

01 StudentCreated
02 StudentEnrolled
03 LessonViewed
04 AssignmentOpened
05 AssignmentSubmitted
06 AssignmentGraded
07 CapabilityObserved
08 CapabilityUpdated
09 RecommendationGenerated
10 PassportUpdated
11 OpportunityMatched
12 EmployerInterested
13 MobilityApproved

```

Appendix C - Example replay request

```

{
  "replay_id": "replay_2026_07_07_001",
  "requested_by": "platform_admin_001",
  "reason": "Rebuild capability graph after ontology version 2.1.0",
  "scope": {
    "tenant_id": "tenant_sa_001",
    "topics": ["syrka.assessment.v1", "syrka.capability.v1"],
    "from": "2026-01-01T00:00:00Z",
    "to": "2026-07-07T00:00:00Z"
  },
  "target_consumer_group": "capability-graph-replay",
  "dry_run": false
}

```

Appendix D - Example audit log

```

{
  "audit_id": "audit_001",
  "event_id": "01JZCEA00000000000000000001",

```

```

"action": "event_consumed",
"consumer": "capability-inference",
"actor": "svc_capability_inference",
"purpose": "capability_inference",
"occurred_at": "2026-07-07T10:15:05Z",
"authorization_decision": "allow",
"policy_version": "policy@2026.07"
}

```

Appendix E - Example dead letter queue record

```

{
  "dlq_id": "dlq_001",
  "event_id": "01JZCEA00000000000000000001",
  "consumer": "capability-inference",
  "topic": "syrka.assessment.v1",
  "attempts": 5,
  "first_failed_at": "2026-07-07T10:15:06Z",
  "last_failed_at": "2026-07-07T10:22:30Z",
  "error_code": "ONTOLOGY_TERM_NOT_FOUND",
  "error_message": "Capability mapping references ontology term unavailable in version 2.1.0",
  "payload_ref": "s3://syrka-dlq/2026/07/07/dlq_001.json",
  "replay_eligible": true
}

```

Appendix F - Example producer pseudocode

```

const event = buildCanonicalEvent({
  eventType: "AssignmentSubmitted",
  version: "1.0.0",
  sourceSystem: "openedx",
  producer: "learning-management-adapter",
  actor: { type: "student", id: studentId },
  subject: { type: "student", id: studentId },
  resource: { type: "assignment", id: assignmentId },
  payload: { submission_id, assignment_id, course_id, submitted_at },
  provenance: { source_event_type: "submission_created", transformer: "openedx-mapper", transformer_version:
"1.4.0" },
  idempotencyKey: openedx:${sourceSubmissionId}:AssignmentSubmitted:1.0.0
});
await validateAndPublish(event);

```

Appendix G - Example consumer pseudocode

```

for await (const event of consumer.subscribe("syrka.assessment.v1")) {
  if (await alreadyProcessed(event.idempotency_key, consumer.name)) continue;
  await db.transaction(async tx => {
    await applyProjection(tx, event);
  });
}

```

```

    await recordCheckpoint(tx, event.event_id, event.idempotency_key);
  });
  await consumer.ack(event);
}

```

Appendix H - Event mapping table summary

Source family	Native language	Syrka canonical language
Open edX	problem_check, score_changed, submission_created	QuizCompleted, AssignmentGraded, AssignmentSubmitted
Canvas	submission_created, quiz_submission_completed	AssignmentSubmitted, QuizCompleted
Moodle	mod_assign_submission_created, mod_quiz_attempt_submitted	AssignmentSubmitted, QuizCompleted
Sakai	assignment submission, assessment submitted	AssignmentSubmitted, QuizCompleted
Chamilo	assignment submitted, exercise completed	AssignmentSubmitted, QuizCompleted
ILIAS	exercise submitted, test finished	AssignmentSubmitted, QuizCompleted
GitHub	push, pull_request, workflow_run	CommitObserved, PullRequestMerged, ProjectCompleted
LinkedIn	profile/experience/certification import	LinkedInProfileLinked, ProfessionalExperienceObserved, CredentialIssued candidate
Research	works, citations, topics	ResearchPublished, ResearchCapabilityMapped
Employer	job, interest, hiring feedback	JobPosted, EmployerInterested, OpportunityAccepted

23. Final Decision

CEA-001 establishes canonical events as the permanent language of Syrka. Every service, connector, AI model, dashboard, database projection, API, and future LMS integration must produce or consume canonical events rather than relying on platform-specific schemas.

The architecture guarantees that any LMS can be replaced without changing Syrka's intelligence layer; every AI capability operates on canonical, provenance-rich evidence; every event is auditable, explainable, replayable, secure, and versioned; and Syrka's strategic intellectual property remains independent of Open edX or any other external platform.